

# INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

## THE SITE

Location Valero Wiserman, TX -- station KSAT, America/Chicago  
Building OSM 509611436  
Valero Wiserman  
Facade gap not applicable -- one building, no second facade  
Weather record 43,809 real hourly records from KSAT  
Plume physics NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

## THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2022-03-25 (crossing)  
Plant limit 21.0 C  
Notice required 3 h  
Forecast skill 0.50 relative to persistence  
Level anchor sensor  
Condenser bank longest facade  
Switch budget 2 mode changes per day  
Minimum dwell 3 h  
Max dew point 15.0 C (Green Grid WP#46 p.6)

## WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 11 of 24 hours with 2 mode changes. The reactive incumbent took 12 hours -- 1 more -- but declared 1 unsafe hour against the agent's 0. 1 hour was safe but still ran chillers, because the schedule could not afford them.

Agent 11 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)  
Incumbent 12 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s)  
1 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

## EVERY HOUR, AND THE REASON FOR IT

| hh | mode | safe | binding | bound  | limit | actual | margin |
|----|------|------|---------|--------|-------|--------|--------|
| 00 | FREE | yes  | -       | 10.000 | 21.0  | 10.000 | 1.070  |
| 01 | FREE | yes  | -       | 9.720  | 21.0  | 9.440  | 0.925  |
| 02 | FREE | yes  | -       | 9.720  | 21.0  | 9.440  | 0.768  |
| 03 | FREE | yes  | -       | 8.335  | 21.0  | 6.670  | 0.632  |
| 04 | FREE | yes  | -       | 8.605  | 21.0  | 7.220  | 0.767  |
| 05 | FREE | yes  | -       | 8.330  | 21.0  | 6.670  | 0.688  |
| 06 | FREE | yes  | -       | 6.950  | 21.0  | 6.670  | 0.598  |
| 07 | FREE | yes  | -       | 7.220  | 21.0  | 6.110  | 0.570  |

|    |            |     |               |        |      |        |       |
|----|------------|-----|---------------|--------|------|--------|-------|
| 08 | FREE       | yes | -             | 10.890 | 21.0 | 11.110 | 1.161 |
| 09 | FREE       | yes | -             | 15.560 | 21.0 | 16.670 | 2.156 |
| 10 | FREE       | yes | -             | 17.780 | 21.0 | 20.560 | 2.124 |
| 11 | mechanical | no  | dry-bulb      | 21.945 | 21.0 | 25.560 | 1.355 |
| 12 | mechanical | no  | dry-bulb      | 25.000 | 21.0 | 27.220 | 1.011 |
| 13 | mechanical | no  | dry-bulb      | 27.505 | 21.0 | 28.890 | 0.998 |
| 14 | mechanical | no  | dry-bulb      | 30.280 | 21.0 | 30.000 | 1.058 |
| 15 | mechanical | no  | dry-bulb      | 30.555 | 21.0 | 30.000 | 0.946 |
| 16 | mechanical | no  | dry-bulb      | 30.835 | 21.0 | 30.000 | 0.938 |
| 17 | mechanical | no  | dry-bulb      | 30.835 | 21.0 | 30.000 | 1.078 |
| 18 | mechanical | no  | dry-bulb      | 29.165 | 21.0 | 28.330 | 0.984 |
| 19 | mechanical | no  | dry-bulb      | 27.220 | 21.0 | 25.000 | 1.331 |
| 20 | mechanical | no  | dry-bulb      | 24.445 | 21.0 | 20.000 | 1.311 |
| 21 | mechanical | no  | dry-bulb      | 23.605 | 21.0 | 19.440 | 1.493 |
| 22 | mechanical | no  | dry-bulb      | 21.665 | 21.0 | 18.890 | 1.228 |
| 23 | mechanical | yes | switch budget | 18.335 | 21.0 | 17.220 | 0.967 |

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin

actual = what the intake temperature turned out to be, from the station record

margin = group-conditional forecast error for that hour of day, plus plume spread

## THE REASONING, HOUR BY HOUR

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### 00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.000 C, which is 11.000 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.070 C, and the margin is measured, not chosen: 1.070 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

### 01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.720 C, which is 11.280 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.926 C, and the margin is measured, not chosen: 0.926 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

### 02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.720 C, which is 11.280 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.768 C, and the margin is measured, not chosen: 0.768 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

### 03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.335 C, which is 12.665 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.632 C, and the margin is measured, not chosen: 0.632 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

### 04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.605 C, which is 12.395 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.767 C, and the margin is measured, not chosen: 0.767 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

### 05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.330 C, which is 12.670 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.688 C, and the margin is measured, not chosen: 0.688 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind

direction is off by the amount FortyGuard's forecast is actually off by.

**06:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 6.950 C, which is 14.050 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.598 C, and the margin is measured, not chosen: 0.598 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**07:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 7.220 C, which is 13.780 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.570 C, and the margin is measured, not chosen: 0.570 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**08:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 10.890 C, which is 10.110 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.161 C, and the margin is measured, not chosen: 1.161 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**09:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 15.560 C, which is 5.440 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 2.156 C, and the margin is measured, not chosen: 2.156 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**10:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 17.780 C, which is 3.220 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 2.124 C, and the margin is measured, not chosen: 2.124 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**11:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 21.945 C against a 21.0 C limit, so it fails by 0.945 C. It would take a limit 0.945 C higher, or a bound 0.945 C tighter, to change this hour.

**12:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 25.000 C against a 21.0 C limit, so it fails by 4.000 C. It would take a limit 4.000 C higher, or a bound 4.000 C tighter, to change this hour.

**13:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 27.505 C against a 21.0 C limit, so it fails by 6.505 C. It would take a limit 6.505 C higher, or a bound 6.505 C tighter, to change this hour.

**14:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 30.280 C against a 21.0 C limit, so it fails by 9.280 C. It would take a limit 9.280 C higher, or a bound 9.280 C tighter, to change this hour.

**15:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 30.555 C against a 21.0 C limit, so it fails by 9.555 C. It would take a limit 9.555 C higher, or a bound 9.555 C tighter, to change this hour.

**16:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 30.835 C against a 21.0 C limit, so it fails by 9.835 C. It would take a limit 9.835 C higher, or a bound 9.835 C tighter, to change this hour.

**17:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 30.835 C against a 21.0 C limit, so it fails by 9.835 C. It would take a limit 9.835 C higher, or a bound 9.835 C tighter, to change this hour.

**18:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 29.165 C against a 21.0 C limit, so it fails by 8.165 C. It would take a limit 8.165 C higher, or a bound 8.165 C tighter, to change this hour.

**19:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 27.220 C against a 21.0 C limit, so it fails by 6.220 C. It would take a limit 6.220 C higher, or a bound 6.220 C tighter, to change this hour.

**20:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 24.445 C against a 21.0 C limit, so it fails by 3.445 C. It would take a limit 3.445 C higher, or a bound 3.445 C tighter, to change this hour.

**21:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 23.605 C against a 21.0 C limit, so it fails by 2.605 C. It would take a limit 2.605 C higher, or a bound 2.605 C tighter, to change this hour.

**22:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 21.665 C against a 21.0 C limit, so it fails by 0.665 C. It would take a limit 0.665 C higher, or a bound 0.665 C tighter, to change this hour.

**23:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.335 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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